

Instructor: Date:

Learner: Regional Branch:

Program: Center:

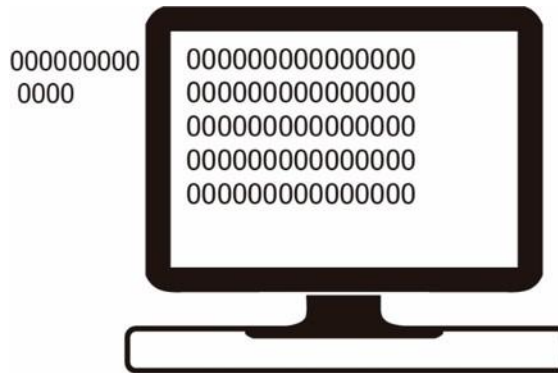
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Let's Start!

The code is the most important documentation in the software development process.



Learning Outcome:

Describe ways of solving problems and getting things done in written and spoken situations and in work and social contexts.

Train the system users on application structuring and management, according to the established plan and user profile, according to the organization policies.

Can Do Statements:

- I can discuss arguments against tracking defects.
- I can talk about the reasons to focus on the software code.
- I can talk about bug reports and their connection to defect tracking.

Ask your partner:

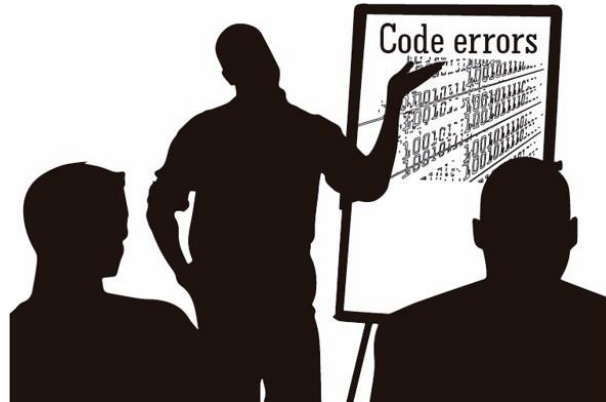
- How do you think defects should be reported? By writing a report, communicating with the person in charge, etc. Why?
- What do you think is the best way to fix bugs?





Let's Explore!

What did the experts say about fixing bugs?



Instructions: Read the text below.

Claire: What did the experts that came last week say about the overall project performance?

Michael: They reached the conclusion that most of our problems were related to the process of fixing bugs. They asked us whether we were tracking all defects throughout the product's life cycle in a very detailed way. Our answer was yes. We thought that the more data we collected on defects, the more we could ensure the end product's quality.

They enquired how our process of fixing bugs was. We told them that once we detected a bug, we wrote a bug report and sent it to the person in charge of fixing it. The bug reports detailed the characteristics of the bug and the formats were designed to be filled out quickly and to provide the necessary information for making the tracking and fixing processes easier.

One of the experts seemed puzzled and asked us if we ever skipped the report's process and made changes directly on the code. Our answer was negative. Our protocol was to always track defects by writing a bug report and only after we had undergone that process, the code could be modified. We decided to ask how direct changes on the code could improve the development process as we thought defect tracking was a great means for collecting data on the source of the defects.

The experts explained that when testing and applying changes directly on the code, we could save time and we would be sure to deliver fixes immediately on the product instead of just waiting for the long defect tracking and fixing processes to be carried out. They added that in this way, the code was always up-to-date and we could make sure that bugs were detected earlier and not pass unnoticed in the documentation.



**Let's Practice!**

Tracking Defects

Instructions: Read the text and select all the sentences about tracking defects.

The experts asked us if we knew that most of our problems were related to the process of tracking defects. Based on that, they asked us whether we were tracking all defects throughout the product's life cycle in a very detailed way. Our answer was yes. They finally enquired whether we thought that the more data we collected on defects, the more we could ensure the end product's quality.

**Let's Practice!**

Bug Reports and Defect Tracking

Instructions: Circle the right answers.

1. The expert (ask / enquire / asked) us why software codes were important.
2. Also, the professor asked (whether / that / what) codes were useful for programming software.
3. The trainer enquired if codes (be / were / was) one of the most important factors when developing software.
4. The engineers asked (mine / yours / us) how codes helped us to modify programming languages
5. They also (said / asked / answered) if codes kept software updated.



Extension Activity

Focusing on Software Code

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Read information about software code.
- Describe how software code works.
- Write a summary of the software code process.
- Show your summary to your classmates and teacher.

Self-Assessment

Yes

No

- I read information about software code.
- I describe a software code process.
- I write a summary software code process.
- I show my summary to the classmates and teacher.

☐	☐
☐	☐
☐	☐
☐	☐



Target Words

- **reach a conclusion** (expression): to end a discussion by agreeing on something, to conclude.
e.g. They reached the conclusion that most of our problems were related to the process of fixing bugs.
- **means** (noun): the way or method for something.
e.g. We decided to ask how direct changes on the code could improve the development process as we thought defect tracking was a great means for collecting data on the source of the defects.
- **up-to-date** (adjective): something that has the latest information.
e.g. They added that in this way, the code was always up-to-date and we could make sure that bugs were detected earlier and not pass unnoticed in the documentation.
- **enquire** (verb): to ask.
e.g. They enquired how our process of fixing bugs was.
- **detail** (verb): to inform by providing a lot of information, to say or write something in a detailed manner.
e.g. The bug reports detailed the characteristics of the bug and the formats were designed to be filled out quickly and to provide the necessary information for making the tracking and fixing processes easier.
- **puzzled** (adjective): confused.
e.g. One of the experts seemed puzzled and asked us if we ever skipped the report's process and made changes directly on the code.



Learning Points

Reported speech in questions

We can use the **reported speech** to report a question someone asked.

For the **reported speech** in questions, just as in the reported speech in statements, the verb tenses in the reported question change depending on the verb that introduces that reported question.

However, we are not going to use the relative pronoun *that* for introducing the reported question but instead, we can use a 'wh' word, or *whether* or *if*.

In the **reported speech in questions**, instead of using verbs like 'say' or 'tell' for introducing the reported question, we are going to use verbs like 'ask' or 'inquire'.

Example:

*The manager asked **what the most common defects** we were finding **were**.*

*The manager asked us **how** we **fixed** bugs.*

The manager asked if we wrote bug reports.

Notice that as the questions are indirect and not direct, we do not place the verb before its subject.

Examples

- *They asked us whether we were tracking all defects throughout the product's life cycle in a very detailed way.*
- *They enquired how our process of fixing bugs was.*
- *One of the experts seemed puzzled and asked us if we ever skipped the report's process and made changes directly on the code.*

Understanding bugs and bug reports

During the software development process, whenever a **bug** is found, it has to be reported in order to get fixed.

A **bug** is any flaw or failure that causes the software to work unexpectedly. These bugs need to get fixed and for doing so, the person that detects the bug has to fill out a bug or defect report and send it to the person in charge of fixing it.

Even though the basic rule for a bug report is that it should explain exactly how a product is broken, not everyone writes bug reports in the same way. The information entered can vary. Here are some tips for writing a good bug report.

- Contain information needed for reproducing and fixing problems.
- Write efficient, precise, and relevant information.
- Add useful detailed information.
- Use a specific format for writing down the report.

